

Re-analysis Summary Report for Peer-evaluation

Paper ID: Christensen_EurJournPersonality_2018_8R9d

Re-analyst's ID: jx7yo

Your task

As a peer-evaluator, for this paper you are asked to judge whether the analyst's applied analytical choices for Task 1 and Task 2 are acceptable. By acceptable, we mean that the analysis pipeline is within the variations that could be considered appropriate by the scientific community in addressing the given analytical task. In addition, for Task 1, we ask you to judge whether the reported conclusion adequately follows from the results of the analysis; whether the co-analyst's self-categorization of the result is adequate. Although, you are not required, it would be appreciated, if you could execute the accompanying analysis code to check any mismatch between the results of the analysis and the reported results.

For both tasks, you should provide your evaluation via this survey: <https://forms.gle/EAREaC6JjN2PWFfR6>

The re-analyst's task

The re-analyst was instructed to assess the following claim of the original authors:

... the high openness to experience group's network was more interconnected ... than the low openness to experience group (p. 480.)

The corresponding article

Here you can find the original article: https://osf.io/6ve3b/?view_only=c874fcf097644e82a320679d4641163a and the original data: https://osf.io/6zh34/?view_only=3e5bfed2e1f041dbbb4376e9c0c5f70b

They are provided only as context of the re-analysis.

In Task 1, the analyst was asked to conduct the analysis without any restrictions. Here, the analyst reported the following steps and results of the analysis:

1. For the data I used the data, published by the authors themselves, as I was agree with the way they did it 2. I calculate the matrix of entropy values between all pairs of unique words based on the frequency of co-occurrences in the individual answers 3. Individual matrices were created on the basis of individual words by choosing appropriate values from the entropy matrix 4. Network measures were calculated for each individual network 5. Correlations and t-test were used to estimate the effect of openness on individual networks

The hypothesis was as follows: higher openness to experience is associated with higher level of interconnections in the individual networks (higher participation coefficient)

Tests suggest that the Spearman correlation between openness and participation coefficient is 0.17 ($p < 0.01$), meaning higher interconnectedness. Comparison of high and low groups (divided by the median of the openness distribution) also shows the higher participation level in the higher group.

The analyst's conclusion: Higher openness to experience is associated with higher participation coefficient

of individual networks

The analyst's categorization of the result:

The results show evidence for the relationship/effect as described in the claim provided in your task

In Task 2, the analysts received the following instruction to conduct the analysis:

In the partial network that you use, the rate of nodes remaining should be 90%.

Here, the analyst reported the following steps and results of the analysis:

I've tested the hypothesis on gender differences in the measures of semantic networks. I assumed that level of participation coefficient should increase with age. It is a simple hypothesis without accounting for difference in openness or age differences. I've constructed individual semantic networks from general network coefficients. Network coefficients were estimated as an entropy of a probability of cooccurrences of two words in one answer. Individual network was constructed by choosing the coefficients of the on the intersection of the words from individual word list. All other coefficients were set to 0.99. The median path length, mean and maximum betweenness centrality and mean participation coefficient were calculated for the MST of each individual network. Gender differences were assessed via Welch two-sample t-test.

Analysis language/software/code (optional to check):

Task 1: R

Task 2: R

Analysis codes: <https://osf.io/2yxqz/>